

Programiz
Online Java Compiler

Main.java

Run

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Clear

```
1- import java.util.ArrayList;
2- import java.util.Iterator;
3- public class Main {
4-     public static void main(String[] args) {
5-         ArrayList<Integer> employeeIds = new ArrayList<>();
6-         employeeIds.add(101);
7-         employeeIds.add(102);
8-         employeeIds.add(103);
9-         employeeIds.add(104);
10-        employeeIds.add(105);
11-        employeeIds.add(106);
12-        employeeIds.add(107);
13-        employeeIds.add(108);
14-        employeeIds.add(109);
15-        employeeIds.add(110);
16-        System.out.println("Original list: " + employeeIds);
17-        Iterator<Integer> it = employeeIds.iterator();
18-        while (it.hasNext()) {
19-            int id = it.next();
20-            if (id % 2 == 0) {
21-                it.remove();
22-            }
23-        }
24-        System.out.println("Updated list (only odd IDs): " + employeeIds);
25-    }
26- }
```

Output

Original list: [101, 102, 103, 104, 105, 106, 107, 108, 109, 110]
Updated list (only odd IDs): [101, 103, 105, 107, 109]

=== Code Execution Successful ===

A college stores the names of students in an ArrayList. The administrator wants to display all student names using an Iterator instead of a for loop.

Write a Java program to:

Create an ArrayList of student names.

Read 8 student names from the user.

Traverse the list using an Iterator.

Display each student name on a separate line.

Display the total number of students stored in the list.

Your Code

```
1 import java.util.ArrayList;
2 import java.util.Iterator;
3 public class Main {
4     public static void main(String[] args) {
5         ArrayList<String> students = new ArrayList<>();
6         students.add("ram");
7         students.add("mohan");
8         students.add("arun");
9         students.add("varun");
10        students.add("manas");
11        students.add("joel");
12        students.add("manish");
13        students.add("yash");
14        System.out.println("student names:");
15        Iterator<String> it = students.iterator();
16        while (it.hasNext()) {
17            System.out.println(it.next());
18        }
19        System.out.println("\ntotal Students = "+students.size());
20    }
21 }
```

Input

stdin input

Output

```
student names:
ram
mohan
arun
varun
manas
```

QUESTIONS

Q1
A college stores the names of students in an ArrayList. The administrator wants to display all student names using an iterator instead of a for loop.
10 marks

Q2
Remove Even Numbers Using Iterator
10 marks

Q3
Product Price Analysis Using Iterator
10 marks

2/3 answered

Q3 Product Price Analysis Using Iterator

10 marks

A supermarket stores the prices of various products in an ArrayList<Double>. The manager wants to analyze the prices using an iterator.

Write a Java program to:

- Create an ArrayList<Double>.
- Read the prices of 10 products from the user.
- Traverse the list using an iterator.
- Calculate and display:
 - Total price of all products.
 - Average product price.
 - Highest product price.
 - Display all product prices using the iterator.

Your Code

```
import java.util.ArrayList;
import java.util.Iterator;

public class Main {
    public static void main(String[] args) {
        ArrayList<Double> prices = new ArrayList<>();
        prices.add(100.0);
        prices.add(250.0);
        prices.add(150.0);
        prices.add(150.0);
        prices.add(400.0);
        prices.add(500.0);
        prices.add(350.0);
        prices.add(450.0);
        prices.add(200.0);
        prices.add(600.0);
        Iterator<Double> it = prices.iterator();
        double total = 0;
        double highest = prices.get(0);
        System.out.println("Product Prices:");
        while (it.hasNext()) {
            double price = it.next();
            System.out.println(price);
            total += price;
            if (price > highest) {
                highest = price;
            }
        }
        double average = total / prices.size();
        System.out.println("Total Price = " + total);
        System.out.println("Average Price = " + average);
        System.out.println("Highest Price = " + highest);
    }
}
```

Input

stdin input

Output

```
100.0
Total Price = 3300.0
Average Price = 330.0
Highest Price = 600.0
```